Tutorial No.7

Q.I] Examine whether the following mapping is Linear Transformation or Not, Justify your answer

1) $T: \mathbb{R}^3 \to \mathbb{R}^4$ **Defined as** $T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x+3z \\ x+y+3z \\ y-x \\ x+z \end{bmatrix}$

2) $T: \mathbb{R}^3 \to \mathbb{R}^1$ **Defined as** $T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = [10x + 20y - 30z]$

3) $T: \mathbb{R}^4 \to \mathbb{R}^3$ **Defined as** $T\left(\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}\right) = \begin{bmatrix} x \\ x+y \\ y+z \\ 0 \end{bmatrix}$

Q.II] Find the matrix of Linear Transformation with respective to standard basis.

1) **If T:** $\mathbb{R}^3 \to \mathbb{R}^4$ **defined as** $T\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x-3y \\ y+3z \\ z+3x \\ x-3z \end{bmatrix}$

2) **If T:** $M_2(R) \to P_3$ **defined as** $T\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ax^3 + bx^2 + cx + d$

3) **If T:** $P_3 \to P_2$ **defined as** $T[p(x)] = \frac{d}{dx}[p(x)]$